

Efficacy of Biopesticides for Management of the Swede Midge (Diptera: Cecidomyiidae)

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Abstract

Management of the swede midge, *Contarinia nasturtii* Kieffer, in North American crucifer production relies on crop rotation and the timely application of synthetic insecticides, based on pheromone trap monitoring of local adult populations. Organically acceptable formulations of azadirachtin, pyrethrin, and spinosad, and a commercial biopesticide containing the entomopathogenic fungus, *Beauveria bassiana*, were evaluated for their effects on larval mortality and oviposition deterrence in the greenhouse, and on damage symptoms in the field. In greenhouse trials, pyrethrin and spinosad treatments applied up to 24 h prior to *C. nasturtii* exposure resulted in significant reductions in oviposition on host plants, whereas azadirachtin and *B. bassiana* only deterred oviposition when applied 2 h prior to exposure. Spinosad caused the highest larval reduction (~96%) on cauliflower meristems, while azadirachtin, *B. bassiana*, and pyrethrin caused significant larval reduction when applied pre-oviposition and significant mortality when applied postoviposition. Field trials conducted with these insecticides on broccoli in 2011 produced no significant reductions in overall damage levels; however, *B. bassiana* treatments produced more marketable plants than did the control. In 2013, all treatments significantly reduced overall damage levels and all treatments, except *B. bassiana*, produced more uninfested and marketable plants than the control. Field applications of these alternative insecticides may be effective in protecting yields of broccoli and cauliflower, when combined with other tactics in an integrated pest management program.

Key words: *Contarinia nasturtii*, pyrethrin, spinosad, azadirachtin, *Beauveria bassiana*

The swede midge, *Contarinia nasturtii* Kieffer (Diptera: Cecidomyiidae), is a serious pest of cultivated vegetables and plants in the family Brassicaceae, including cultivars of *Brassica oleracea* L., such as broccoli, cauliflower and cabbage, cruciferous weeds (e.g., *Sinapis arvensis* L., *Raphanus raphanistrum* L., and *Capsella bursa-pastoris* L.), and canola (*B. napus* L., *B. juncea* (L.) Czern., and *B. rapa* L.) (Hallett 2007, Chen et al. 2011). Individual female *C. nasturtii* can lay several clutches of up to 50 eggs within the tightly folded leaves surrounding the host plant meristem and developing buds (Barnes 1946). Larvae feed by extraintestinal digestion, using salivary enzymes to break down the meristematic cells of the host plant, resulting in a wide range of symptoms, including twisted and swollen stems, crumpled leaves, multiple heads, or the complete absence of the vegetable head (Barnes 1946, Hallett and Heal 2001). This direct damage to the harvestable portion of the plant can lead to significant yield reductions, with losses of up to 85% reported in heavily infested vegetable fields (Hallett and Heal 2001).

Contarinia nasturtii larvae develop to maturity on the host plant, dropping into the soil to spin a cocoon and pupate before emerging as adults. Life cycle duration is variable, ranging from 20–55 d, depending on climatic variables such as temperature and humidity (Readshaw 1966). *Contarinia nasturtii* is a multivoltine species that

exhibits considerable intragenerational variation in adult emergence patterns, characterized by temporally distinct early- and late-emerging phenotypes (Hallett et al. 2009b). As the growing season progresses, the staggered emergence cohorts cause sequential generations to overlap, resulting in the concurrent presence of all life stages of *C. nasturtii* in the field (Hallett et al. 2009b, Chen et al. 2011).

The presence of *C. nasturtii* in North America was first confirmed in Ontario, Canada, in June 2000 (Hallett and Heal 2001). Since that time, its North American range has expanded rapidly into the Canadian provinces of Quebec, Nova Scotia, Prince Edward Island, Manitoba, and Saskatchewan and into the states of New York, New Jersey, Massachusetts, Connecticut, Vermont, Ohio, and Michigan in the United States (Chen et al. 2007, 2009; Canadian Food Inspection Agency [CFIA] 2009, Government of Michigan 2015). In response to this rapid range expansion, the Canadian Food Inspection Agency and the United States Department of Agriculture Animal Plant Health Inspection Agency abandoned efforts to regulate *C. nasturtii* as a quarantine pest in 2009 (CFIA 2009).

Researchers focusing on the efficacy of synthetic insecticides identified foliar sprays (acephate, acetamiprid, chlorpyrifos, λ -cyhalothrin, dimethoate, methomyl, and permethrin), soil drenches (acetamiprid, imidacloprid, and thiamethoxam), and seed treatments

(clothianidin and thiamethoxam) that could effectively reduce feeding damage by *C. nasturtii* (Wu et al. 2006, Hallett et al. 2009b). However, field trials conducted in 2005 and 2006 showed that insecticide treatments that had proven effective during the early stages of *C. nasturtii* establishment in North America (2001 and 2002) subsequently failed to reduce feeding damage to levels acceptable for marketability (Hallett et al. 2009b). It has been postulated that the multiple, overlapping generations of *C. nasturtii*, combined with short residual activity of insecticides, are the leading causes of control failures in North America, rather than the development of insecticide resistance (Chen et al. 2007, 2011). The inconsistent effectiveness of synthetic insecticides, the complex life history traits of *C. nasturtii*, and the absence of endemic, specialized natural enemies in North America have led to the conclusion that an integrated pest management approach is critical to maintain *C. nasturtii* populations at economically acceptable levels (Corlay et al. 2007, Hallett et al. 2009b, Chen et al. 2011, Abram et al. 2012).

Current management efforts by North American crucifer producers involve biannual crop rotations and the application of synthetic insecticides in response to *C. nasturtii* infestations, incorporating the use of insecticide application thresholds based on pheromone trap monitoring of adult males (Hallett et al. 2007, Chen et al. 2011, Hallett and Sears 2013). The successful management of *C. nasturtii* in organic production systems requires the identification of effective alternative insecticides. Biopesticides are pesticides derived from natural materials, such as animals, plants, and bacteria, which may be commercially produced with synthetic additives for use by conventional growers, or without synthetic additives for organic producers (US Environmental Protection Agency [USEPA] 2016). Organically acceptable biopesticides provide management tools to both organic and conventional growers, offer the benefit of different modes of action, and reduce the potential for resistance development, which will enhance long-term management of swede midge. The objective of this research was to examine the effect of commercially available alternative insecticide formulations on *C. nasturtii* rates of oviposition, larval mortality, and feeding damage on cole crops.

Materials and Methods

Insect Culture

Contarinia nasturtii larvae from the Swiss Federal Research Station for Horticulture (Wädenswil, Switzerland) were obtained from Dr. Anthony Shelton (Cornell University, Geneva, NY) in 2011 and reared following the methods described in Chen and Shelton (2007) and Des Marteaux et al. (2012). Adults and larvae were kept at $25 \pm 1^\circ\text{C}$, $60 \pm 10\%$ RH, and a photoperiod of 16:8 (L:D) h.

Plant Material

For both colony maintenance and experiments, 'Snow Crown' variety cauliflower (*Brassica oleracea* L.) (Stokes Seeds, Thorold, Ontario, Canada) was seeded into 128-cell flats with Promix (Premier Horticulture Ltd., Dorval, Québec, Canada) and grown in a greenhouse at the University of Guelph, Guelph, Ontario, Canada. Weekly fertilizer applications were made to seedling flats using 10-52-10 "plant starter" fertilizer (Plant Products Co. Ltd., Brampton, Ontario, Canada) and after 6–8 wk, seedlings were transplanted, in pairs for colony or singly for experiments, into 13-cm-diameter pots and fertilized weekly with 20-20-20 "all purpose" fertilizer (Plant Products Co. Ltd., Brampton, Ontario, Canada). Potted plants were introduced to colony oviposition cages or used for experiments at the 8–10 true leaf stage, when the circumference of the apical meristem exceeded 5 cm.

Insecticides

The active ingredients and products examined herein are either permitted organic substances in Canada (Canadian General Standards Board 2015) that are regulated by the Canadian Pest Management Regulatory Agency, or are OMRI Listed products (Organic Materials Review Institute, www.omri.org). Treatments for all greenhouse and field experiments consisted of pyrethrin (Pyganic EC 1.4, 1.4% pyrethrins (w/w), McLaughlin Gormley King Company, Minneapolis, MN), spinosad (Entrust 80W, 80% spinosad (w/w), Dow AgroSciences LLC, Calgary, Alberta, Canada), azadirachtin (Leaf Shine, pure neem oil with 2,000 ppm azadirachtin, Biofert Manufacturing Inc., Langley, British Columbia, Canada), and *Beauveria bassiana* (Botanigard 22WP, *B. bassiana* Strain GHA, 4.4×10^{13} conidia/kg, Laverlam International Corporation, Butte, MT), and distilled water as a control. For all experiments, the application rates recommended by the manufacturers were used (i.e., pyrethrin, 60.9 g ai/ha; spinosad, 87.4 g ai/ha; *B. bassiana*, 1,076 g ai/ha; and azadirachtin, 13.5 g ai/ha) and applied with the equivalent of 350 liter of water per hectare. Application rates to individual plants in laboratory experiments were based on a density of ~29,000 plants/ha.

Oviposition Deterrence Experiments

Methods for both oviposition deterrence and larval mortality experiments (described below) were informed by those of Wu et al. (2006) and Chen and Shelton (2010). Two separate greenhouse experiments were conducted to evaluate the effects of insecticide treatments on *C. nasturtii* oviposition. Individual plants were treated with one of the four insecticides or a distilled water control. Treated plants were placed individually inside 25- by 25- by 25-cm Bugdorm wire-framed rearing and observation cages (Megaview, Taiwan), with plastic covering the front and fine mesh covering the top, bottom, and three sides, and maintained at $25 \pm 1^\circ\text{C}$. In the first experiment, 10 newly emerged male and 10 newly emerged female adult *C. nasturtii* (24–48 h old), having had no exposure to host plants and, thus, no opportunity to lay eggs, were collected from colony emergence cages using an aspirator and immediately introduced into a cage 2 h following the application of insecticides. The second experiment followed the same protocol, except that adult midges were introduced to the experimental cages 24 h following the application of insecticides. In both experiments, insecticides were applied in ~10 ml water/plant directly to the foliage using a misting bottle held ~30 cm above the plant, immediately prior to placing plants in experimental cages. Plants were not watered for the duration of the experiments (~72 h). Each treatment was replicated five times and each experiment was conducted twice. Cages were placed on lab benches and arranged in a completely randomized design. Three days following the introduction of adult midges into the experimental cages, plants were removed and the apical meristem and surrounding leaves were dissected and examined under a dissecting microscope. All eggs and larvae found were counted.

Larval Mortality Experiments

Greenhouse experiments were conducted to evaluate the effects of insecticide treatments on larval mortality. Plants were treated in the same way as the oviposition deterrence experiments. In a preoviposition treatment experiment, 10 male and 10 female newly emerged *C. nasturtii* were introduced into each experimental cage containing an individual cauliflower plant that was treated with insecticide 24 h earlier.

A second, postoviposition treatment experiment followed the same protocol, except that all plants were exposed to midges for 72 h, allowing oviposition and early larval development to occur, prior to the application of insecticide treatments. In this experiment, plants were removed from experimental cages, treated with insecticide, and immediately placed back into the cages. At the time of treatment, nearly all adult *C. nasturtii* in the experimental cages had died naturally, either on the plant foliage or at the bottom of the cages.

For both experiments, each treatment was replicated five times and each experiment was conducted twice. Cages were placed on lab benches and arranged in a completely randomized design. Approximately 150 ml of water was applied to the soil at the base of the potted plants on the third or fourth day of the experiment. Plants were removed from the experimental cages 7–8 d following the introduction of adults (when larvae were ~4 d old) and apical meristems and surrounding leaves were examined and dissected under the microscope. All larvae were removed from the plant material with fine paint brushes and recorded as dead or alive, with the proportion of dead:total larvae recovered expressed as percent larval mortality. In the preoviposition treatment trial, total larval reduction was also determined for each treatment in order to ascribe proportions of the reduction in number to either oviposition deterrence or larvicidal activity. Percent reduction in live larvae per plant was calculated as:

$$\% \text{Live larval reduction} = \sum_{i=1}^n \left(\left[\frac{L_{ci} - L_{ti}}{L_{ci}} \right] / n \right) \times 100$$

where;

L_{ci} = number of live larvae on a control plant for a given replicate,

L_{ti} = number of live larvae on a plant treated with treatment “t” for a given replicate, and

n = number of replicates.

In the preoviposition experiment, oviposition deterrence could not be determined based on counting the number of eggs on the meristem, since this form of sampling would require dissection of the host plant and preclude any further determination of larvicidal activity. Rather, percent oviposition deterrence was inferred based on the reduction in total larvae present on the meristem, calculated as:

$$\% \text{Oviposition deterrence} = \sum_{i=1}^n \left(\left[\frac{T_{ci} - T_{ti}}{T_{ci}} \right] / n \right) \times 100$$

where;

T_{ci} = number of total (live + dead) larvae on a control plant for a given replicate, and

T_{ti} = number of total larvae on a plant treated with treatment “t” for a given replicate.

The proportion of the overall larval reduction attributable to oviposition deterrence was then calculated as:

$$P(O) = \sum_{i=1}^n \left[\frac{D_{ti}}{R_{ti}} \right]$$

and, the proportion of the reduction attributable to larvicidal activity was calculated as:

$$P(L) = \sum_{i=1}^n \left[\frac{K_{ti}}{R_{ti}} \right], \text{ or } = 1 - P(O)$$

where;

D_{ti} = number of larvae deterred as a result of the presence of the biopesticide for a given replicate, calculated as $D_{ti} = T_{ci} - T_{ti}$, and

K_{ti} = number of larvae killed on the meristem due to larvicidal activity for a given replicate, calculated as $K_{ti} = T_{ti} - L_{ti}$, and

R_{ti} = reduction in overall larval presence on the meristem for a given replicate, calculated as $R_{ti} = K_{ti} + D_{ti}$.

Insecticide Efficacy Field Trials

Field experiments were conducted during the months of July and August in Zephyr, ON, in 2011 and Elora, ON, in 2013. Trials were conducted with ‘Windsor’ in 2011 and with both ‘Windsor’ and ‘Bay Meadows’ broccoli (Stokes Seeds Ltd., Thorold, Ontario, Canada) in 2013. All seeds were planted as described above and grown in a greenhouse for 4 wk, then moved outside for 7 d prior to transplanting in the field (Zephyr, July 11, 2011; Elora, July 25, 2013). Field plots each consisted of 4 rows of broccoli, 5 m in length, with 45 cm between plants and 1 m between rows (~45 plants), with 2 m buffer spaces between plots. Plots were arranged in a randomized complete block design and replicated four times. Every 7–10 d, for a total of 7 applications at Zephyr and 8 applications at Elora, plants were treated with an insecticide or a deionized water control, at the concentrations described above using a CO₂ backpack sprayer with a one-row boom equipped with TeeJet XR8002VS nozzles. Each week, 10 plants were randomly selected from the two middle rows of each plot and assigned a damage rating, where 0 = no damage, 1 = mild crumpling of leaves, 2 = severe crumpling and twisted stem, 3 = death of apical meristem and/or multiple compensatory shoots (Hallett 2007). Final damage ratings were taken at harvest (Zephyr, August 30, 2011; Elora, September 25, 2013).

Pheromone trap monitoring was conducted at Elora, from May to October during both years of the study (Fig. 1) placing six white Jackson traps baited with *C. nasturtii* sex pheromone polyethylene cap lures (PheroNet Swede Midge Lures, Andermatt Biocontrol, Grossdietwil, Switzerland), ~150 m apart around the perimeter of the experimental field, attached to wood stakes ~30 cm above the ground. Traps were checked three times per week for captured *C. nasturtii* adult males and lures were replaced every 4 wk.

Statistical Analysis

All statistical analyses were conducted with SAS 9.3 (SAS Institute, Cary, NC 2012), and $\alpha = 0.05$. Oviposition deterrence data were analyzed using a generalized linear mixed model (PROC GLIMMIX), assuming a negative binomial distribution and log link function. The resulting means were separated using the Tukey–Kramer procedure. Larval mortality data were arcsine-transformed to produce normally distributed data and treatment effects were analyzed by analysis of variance (ANOVA) using the general linear model procedure (PROC GLM), followed by multiple means comparisons using Tukey’s adjustment for LS Means. No trial by treatment interactions were found for the two trial repetitions of the oviposition deterrence (2 h: $F = 1.58$, $P = 0.23$; 24 h: $F = 1.94$, $P = 0.15$) and larval mortality (preoviposition: $F = 1.57$, $P = 0.20$; postoviposition: $F = 1.38$, $P = 0.26$) experiments, so data were pooled.

For the field trials, in order to compare the distribution of plants among damage rating categories estimated probabilities of damage rating scores at harvest were generated from damage rating data by way of a multinomial analysis using PROC GLIMMIX and a cumulative logit link function. The distributions of the resulting estimated probabilities were then compared using linear contrasts. The mean number of marketable plants (i.e., at harvest damage rating ≤ 1) and uninfested plants (damage rating = 0) were compared across treatments using PROC GLIMMIX, assuming a binomial distribution

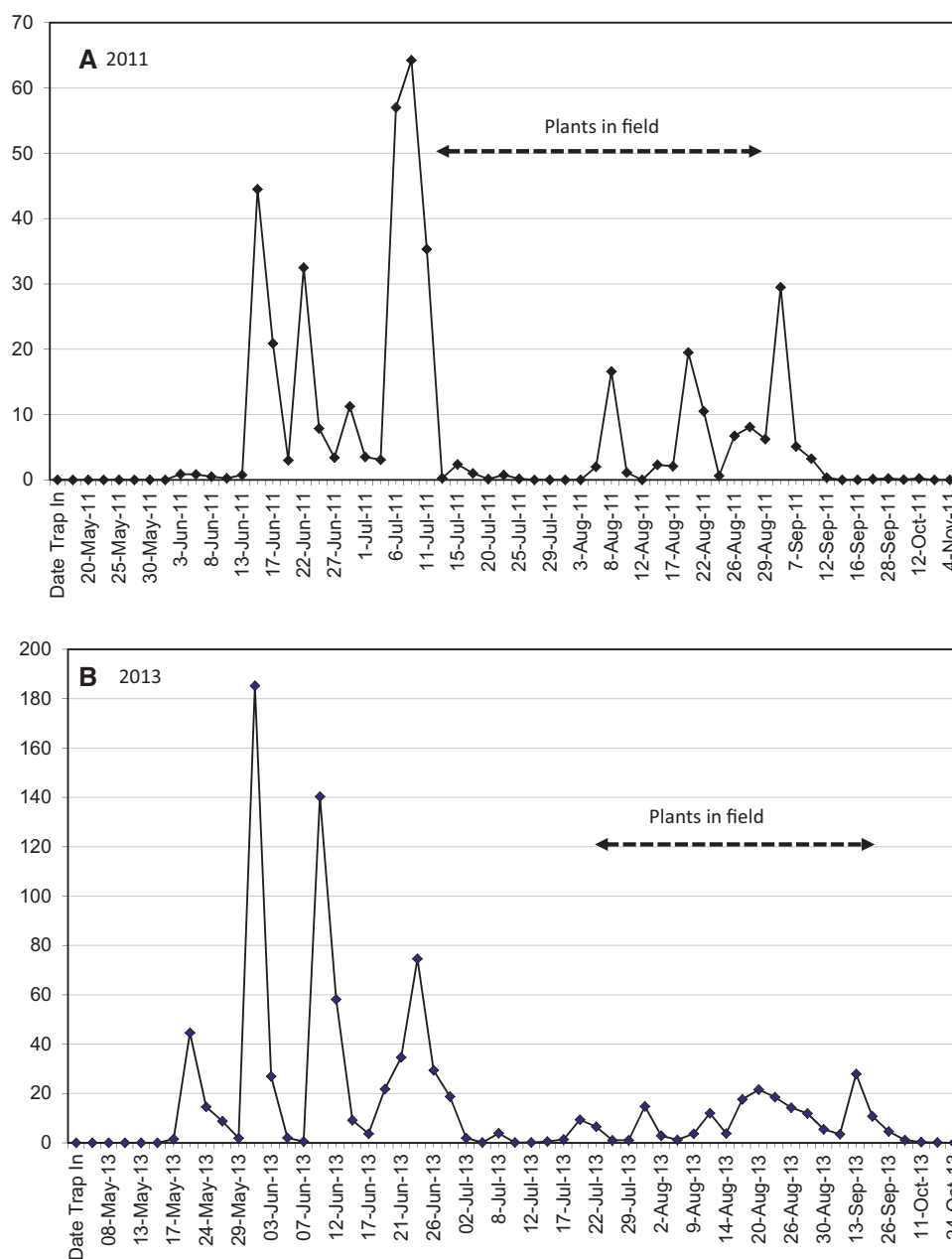


Fig. 1. Mean daily captures (males/trap/day) of *C. nasturtii* adult males in pheromone traps at Elora, Ontario, Canada, (A) 2011 and (B) 2013. Note differences in scales of y-axes.

and logit link function. In addition, mean damage ratings at harvest for each treatment were also compared using PROC GLM. The resulting means were separated using the Tukey–Kramer procedure.

Results

Oviposition Deterrence Experiments

Insecticide treatments made 2 h prior to midge exposure resulted in significantly lower numbers of eggs on plants for all treatments compared to the control ($F = 32.75$, $df = 4, 16$, $P < 0.0001$; Table 1). The largest reduction in oviposition was observed in the pyrethrin treatment (92%), followed by spinosad (87%), and azadirachtin and *B. bassiana* (both 81%). Significant reductions in oviposition also resulted when plants were treated with insecticides 24 h prior to

midge exposure ($F = 14.06$, $df = 4, 16$, $P < 0.0001$; Table 1). Oviposition was significantly reduced by pyrethrin (69%) and spinosad (50%), but not by azadirachtin and *B. bassiana* (both 26%), when treatments were made 24 h before midge exposure.

Larval Mortality Experiments

Significant mortality was observed for all treatments in the preoviposition application trial ($F = 5.45$, $df = 4, 16$, $P < 0.0038$), with spinosad causing the highest level of mortality, followed by *B. bassiana* and azadirachtin (Table 2). The total reduction of live larvae on the meristem by spinosad and pyrethrin was significantly higher than that of azadirachtin and *B. bassiana*, though all treatments greatly reduced larvae ($\geq 76\%$), with a very high proportion of the reduction attributable to oviposition deterrence for all treatments ($\sim 70\text{--}93\%$), as

Table 1. *C. nasturtii* eggs collected from the meristematic tissue of cauliflower following oviposition by adults exposed to plants 2 h and 24 h after application of insecticide treatments

Treatment	No. of eggs (mean $\pm$ SE)	
	Exposure 2 h postapplication	Exposure 24 h postapplication
Azadirachtin	39.2 $\pm$ 6.2b	112.1 $\pm$ 12.1ab
<i>B. bassiana</i>	39.6 $\pm$ 7.9b	112.1 $\pm$ 14.0ab
Pyrethrin	17.0 $\pm$ 5.0c	46.9 $\pm$ 8.1d
Spinosad	26.6 $\pm$ 4.5b	76.3 $\pm$ 11.2c
Control	209.6 $\pm$ 34.7a	151.7 $\pm$ 13.7a

Data pooled for two repetitions of each experiment.

Means within a column followed by the same letter are not significantly different (Tukey HSD, $P > 0.05$).

compared to larvicidal activity (~ 7 – 29%). All postoviposition insecticide treatments resulted in significant mortality compared to the control ($F = 14.89$, $df = 4,16$, $P < 0.0001$; Table 2). Spinosad caused the highest level of mortality, followed by pyrethrin, azadirachtin, and *B. bassiana*.

Insecticide Efficacy Field Trials

Statistical determination of estimated probabilities of damage ratings showed that *C. nasturtii* damage was not reduced by any insecticide treatment at Zephyr ($F = 1.98$, $df = 4,15$, $P > 0.15$; Table 3). At Elora, treatment effects were statistically significant ($F = 9.00$, $df = 4,15$, $P = 0.0047$), but no significant varietal effect ($F = 4.55$, $df = 1,2$, $P > 0.16$) nor treatment by variety interaction ($F = 0.09$, $df = 4,8$, $P > 0.98$) was observed, and so data from both varieties were pooled for analyses of main treatment effects. At Elora, all insecticide treatments were found to differ significantly from the control in their estimated probabilities of damage rating distributions, and all had higher probabilities of having plants with no damage, and lower probabilities of having plants with no head, than did the control (Table 3).

At Zephyr, 75% of plants in the *B. bassiana* treatment were rated marketable at harvest, which was significantly higher than the control (Table 4). However, no differences in the mean number of uninfested plants were observed among treatments in this trial. At Elora, there were significant differences among treatments in the number of marketable plants at harvest (Table 4), but there were no significant variety ($F = 4.96$, $df = 1,2$, $P = 0.16$) nor variety by treatment interaction ($F = 0.35$, $df = 4,8$, $P = 0.84$) effects, so varietal data were pooled for analyses. All insecticides, except *B. bassiana*, had significantly more marketable plants than the control. Varietal data for the number of uninfested plants were also pooled for analyses (variety: $F = 8.34$, $df = 1,2$, $P = 0.10$; variety by treatment interaction: $F = 0.35$, $df = 4,8$, $P = 0.84$). All insecticides, except *B. bassiana*, also had more uninfested plants than did the control.

Patterns of adult emergence observed at Elora in 2011 (Fig. 1) may serve as a reasonable estimate of the timing (though not magnitude) of emergence events at Zephyr, given their geographic proximity (Zephyr, 44.1167° N, 79.1333° W; Elora, 43.6850° N, 80.4272° W) and comparable climates (average daily maximum temperatures for June and July, 2011: Zephyr, 23.7°C and 28.6°C, recorded at Udora, 44.1545° N, 79.0941° W; Elora, 22.4°C and 28.0°C, recorded at Fergus Shand Dam 43.4405° N, 80.1949° W; Government of Canada, 2015). Pheromone trap data indicate that transplanting of seedlings at Zephyr occurred toward the end of the

largest emergence peak of the 2011 growing season, while transplanting at Elora occurred ~ 4 wk after the last large early season emergence peak in 2013.

At Zephyr and Elora, initial damage levels developed at a similar rate, with damage levels very low or absent during the first 4 wk after transplanting, followed by steady increases in observed damage from the fifth week onward (Fig. 2). Damage ratings for all treatments increased similarly over time at Zephyr, while at Elora damage in the control began to diverge from that of the insecticide treatments at 5 wk after transplanting. At harvest, *B. bassiana* and pyrethrin had significantly lower damage ratings than the control at Zephyr ($F = 6.59$, $df = 4,16$, $P < 0.0001$), while all treatments had significantly lower damage ratings than the control at Elora ($F = 14.8$, $df = 4,16$, $P < 0.0001$; Fig. 2). Mean damage levels in the controls reached similar levels at both sites (~ 2.0).

Discussion

All insecticides evaluated herein exhibited a very high degree of oviposition deterrence when applied 2 h prior to exposure to gravid females; however, spinosad and pyrethrin also significantly reduced oviposition for at least 24 h after application. The higher egg counts observed on plants treated with *B. bassiana* and azadirachtin, particularly 24 h postapplication, suggest that any deterrent effect of these active ingredients on ovipositing females is very short-lived. Although azadirachtin is purported to have strong repellent effects on several phytophagous insect species (Kumar and Poehling 2006), the relatively low concentration used in our experiments, and rapid breakdown under UV radiation may explain its relative ineffectiveness in our experiments (Dayan et al. 2009). The selection of a suitable host plant for oviposition is of critical importance for insect species where the larval stage exhibits low mobility (Messina et al. 1987, Renwick and Chew 1994), so preventing oviposition may be a useful tactic in reducing plant damage by such herbivore pests. *Contarinia nasturtii* larvae lack the ability to move from one plant to another and, thus, their survival and successful development to the adult stage is dependent on the suitability of the oviposition site selected by the female (Readshaw 1966, Huang et al. 1994). The results suggest that well-timed applications of pyrethrin and spinosad could have a deterrent effect on gravid females seeking oviposition sites for at least 24 h, helping to reduce eventual larval numbers and augmenting any larvicidal effect of the residual insecticide on the meristem. This is the first report of oviposition deterrent effects of insecticides on *C. nasturtii*.

Contarinia nasturtii larvae may spend as few as 7 d feeding on the host plant in order to complete larval development and, since the growing points of plants are attacked in the early stages of development, damage caused by the larvae can quickly result in the loss of the harvestable portion of the plant (Hallett and Sears 2013). Marketability of some host plants, such as cabbage and kale (*B. oleraceae* L.), may be minimally affected by later damage, once the meristem is protected from ovipositing females by head formation and/or since damaged older leaves are removed at harvest. However, cauliflower and broccoli are vulnerable to direct damage to developing florets, and may only remain marketable by minimizing damage by *C. nasturtii* throughout the full growing season (Hallett 2007). Biopesticide treatments aimed at reducing the opportunities for *C. nasturtii* oviposition on untreated host plants in susceptible stages of development may be achieved via population monitoring by pheromone trap captures and timing applications to coincide with emergence peaks. Foliar applications of synthetic

Table 2. Percent larval reduction, larval mortality, and attribution of reduction to oviposition deterrence or larvicidal activity, in the meristematic tissue of cauliflower treated with insecticide before and after oviposition

Treatment	Preoviposition application						Postoviposition application
	Total larvae/plant	% Larval mortality	% Oviposition deterrence	% Live larval reduction	P(O) ^a	P(L) ^b	% Larval mortality
	(mean ± SE)	(mean ± SE)	(mean ± SE)	(mean ± SE)			(mean ± SE)
Azadirachtin	35.7 ± 5.6	31.7 ± 6.8b	78.7 ± 0.05a	84.0 ± 0.05ab	0.930	0.70	46.7 ± 11.0ab
<i>B. bassiana</i>	45.4 ± 6.8	31.9 ± 6.7b	68.7 ± 0.08a	75.8 ± 0.07b	0.706	0.294	44.3 ± 8.0b
Pyrethrin	30.6 ± 3.0	56.4 ± 6.3ab	81.4 ± 0.03a	91.2 ± 0.02a	0.873	0.127	62.2 ± 6.7ab
Spinosad	37.8 ± 8.2	78.3 ± 8.4a	78.3 ± 0.04a	95.1 ± 0.02a	0.767	0.233	76.3 ± 7.2a
Control	194.1 ± 27.8	4.2 ± 0.8c	—	—	—	—	5.8 ± 0.8c

^a Proportion of larval reduction attributable to oviposition deterrence.^b Proportion of larval reduction attributable to larvicidal activity.Means within a column followed by the same letter are not significantly different (Tukey HSD, $P > 0.05$).**Table 3.** Estimated probabilities of larval *C. nasturtii* feeding damage on broccoli treated with insecticides in field trials at Zephyr in 2011 and Elora in 2013, where a rating of 0 = no damage, 1 = mild crumpling of leaves, 2 = severe crumpling & twisted stem, 3 = complete absence of head

Location, year, treatments	Estimated probabilities of feeding damage ratings				
	Rating = 0	Rating = 1	Rating = 2	Rating = 3	Pr > F
Zephyr, 2011					
Azadirachtin	0.060	0.364	0.428	0.149	—
<i>B. bassiana</i>	0.187	0.540	0.227	0.046	—
Pyrethrin	0.103	0.467	0.342	0.089	—
Spinosad	0.057	0.354	0.433	0.156	—
Control	0.029	0.229	0.472	0.271	—
Elora, 2013					
Azadirachtin	0.637	0.109	0.037	0.217	0.0007
<i>B. bassiana</i>	0.560	0.120	0.043	0.277	0.0132
Pyrethrin	0.707	0.094	0.031	0.168	0.0005
Spinosad	0.800	0.070	0.022	0.108	0.0024
Control	0.259	0.110	0.049	0.582	—

 P -values are given for statistically significant treatment effects as compared to controls by linear contrasts ($P < 0.05$).**Table 4.** Mean number of uninfested (rating = 0) and marketable (rating ≤ 1) broccoli plants per 10-plant sample following treatment with insecticides in field trials at Zephyr in 2011 and Elora in 2013

Trial location, year, treatments	Mean marketable plants ± SE	Mean uninfested plants ± SE
Zephyr, 2011		
Azadirachtin	3.5 ± 0.9ab	1.5 ± 0.6a
<i>B. bassiana</i>	7.5 ± 1.2a	1.5 ± 0.9a
Pyrethrin	6.0 ± 0.4ab	0.5 ± 0.5a
Spinosad	4.8 ± 0.8ab	0.8 ± 0.5a
Control	2.5 ± 1.0b	0.5 ± 0.5a
Elora, 2013		
Azadirachtin	7.7 ± 1.1a	6.3 ± 1.0a
<i>B. bassiana</i>	6.7 ± 0.9ab	5.7 ± 1.3ab
Pyrethrin	7.7 ± 0.8a	7.2 ± 1.1a
Spinosad	8.8 ± 0.7a	8.0 ± 1.1a
Control	3.7 ± 1.4b	2.3 ± 1.0b

Means within a column followed by the same letter are not significantly different (Tukey HSD, $P > 0.05$).

insecticides have been shown to reduce damage by *C. nasturtii* on host plants in both laboratory and field studies (Wu et al. 2006, Hallett et al. 2009a,b). All alternative products tested caused significant larval mortality in preoviposition and postoviposition trials; however, the organic formulations of spinosad and pyrethrin were most effective, achieving more than twice the level of mortality caused by azadirachtin and *B. bassiana* in preoviposition trials and ~1.5 times of that in postoviposition trials. The mortality levels caused by spinosad (76 and 78%) and pyrethrin (56 and 62%) were also more consistent between the pre- and postoviposition trials than azadirachtin (32 and 47%) and *B. bassiana* (32 and 44%), suggesting that the former pair of treatments may not degrade as quickly on foliage than the latter, leading to longer residual activity. Although all of the organically acceptable insecticides examined herein demonstrated larvicidal properties when applied pre- or postoviposition, they were much less effective than synthetic insecticides evaluated in a similar study by Wu et al. (2006). However, it should be noted that our use of water as a control could have resulted in an underestimation of the efficacy of insecticides compared to other studies where an untreated control was utilized. Postoviposition foliar applications of conventional formulations of the pyrethroid λ -cyhalothrin and spinosad, both of which were applied with a silicone-polyether copolymer adjuvant to improve coverage, caused larval mortality of 97% and 92%, respectively (Wu et al. 2006), as compared to mean mortality of 56% for organic pyrethrin and 78% for organic spinosad in this study.

While larval mortality studies involving postoviposition applications report on the lethality of insecticide treatments, preoviposition application experiments account for overall reductions in larval numbers, attributable to both oviposition deterrence effects and larvicidal effects. The reductions in larvae observed for all treatments in the preoviposition trials were largely attributable to lower numbers of larvae present on plants, i.e., oviposition deterrence (71–93%), rather than to reduction by larvicidal activity (7–29%). Although oviposition deterrence may be overestimated here, since our methods did not account for potential ovicidal activity, this laboratory result suggests that timing of insecticide applications could be manipulated to take advantage of preventative effects, rather than curative effects alone, and that this approach may be particularly important in improving overall efficacy of organic insecticides that cause lower larval mortality than synthetic pesticides.

This study is the first to evaluate efficacy of multiple organic insecticides against *C. nasturtii* in a Canadian field setting. In addition

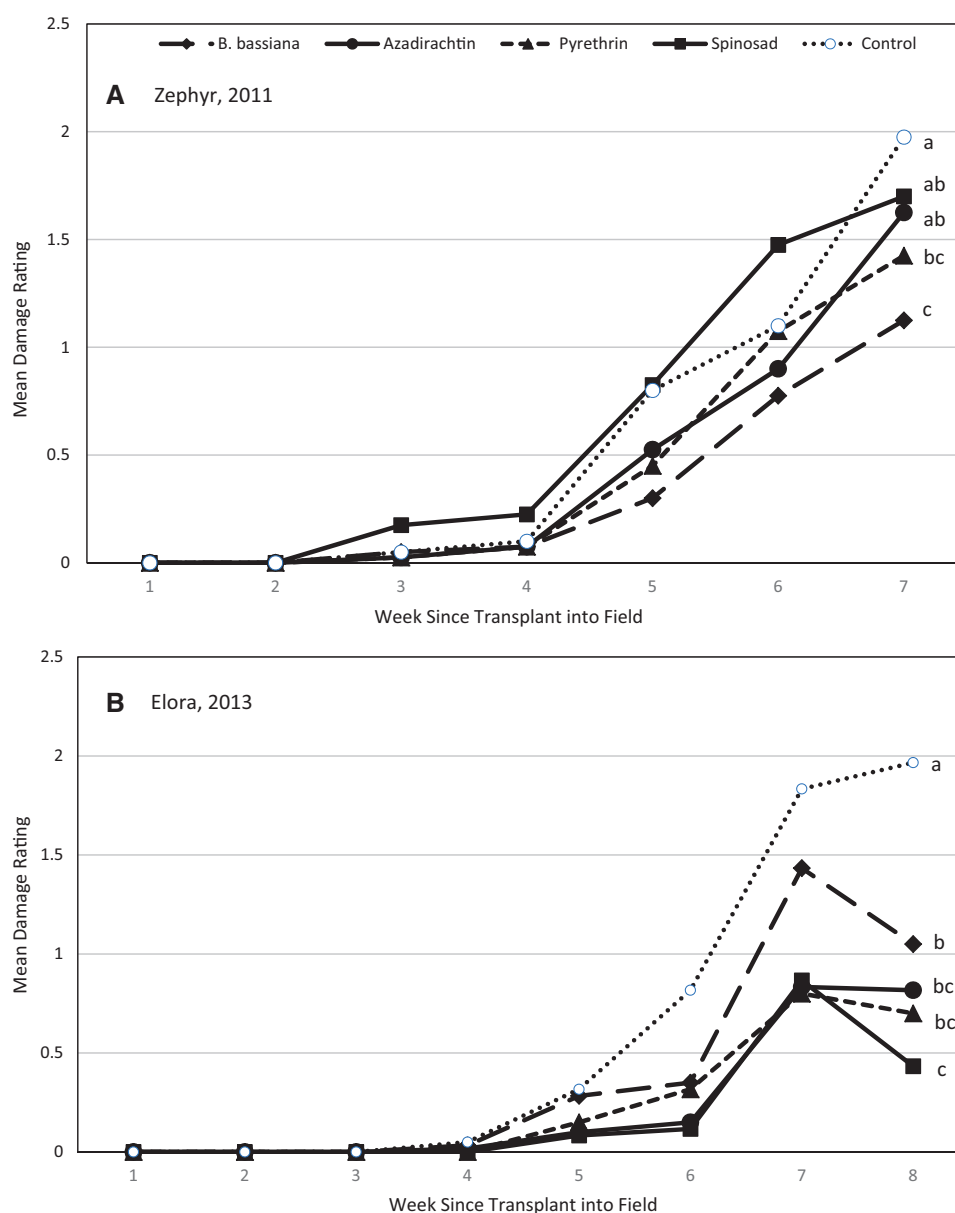


Fig. 2. Mean weekly ratings of *C. nasturtii* feeding damage on broccoli at (A) Zephyr (2011) and (B) Elora (2013), Ontario, Canada, where a rating of 0 = no damage, 1 = mild crumpling of leaves, 2 = severe crumpling & twisted stem, 3 = complete absence of head.

to presenting a comparison of final mean damage ratings analyzed by ANOVA (GLM), as has been done in previous studies using *C. nasturtii* damage ratings (e.g., Hallett et al. 2009a), we elected to utilize a generalized linear mixed model (GLIMMIX), which can be constructed to account for the multinomial distribution of these data (Gbur et al. 2012). Further, the resulting output of estimated probabilities indicates the proportional odds of each damage category for a given treatment. This output provides considerably more information than a singular mean value, which may result from observations that are not normally distributed. While the multinomial analysis (GLIMMIX) showed no significant differences among treatments in the Zephyr (2011) field experiment, the ANOVA (GLM) produced statistically significant results. In reviewing the estimated probabilities across damage rating categories (Table 3), some data are normally distributed about the mean (e.g., *B. bassiana*, 2011), while some other observations are not (e.g., control, 2013). The multinomial approach conducted herein thus better fits the

distribution of these data, and so should provide a more reliable assessment of the relative efficacies of insecticide treatments than does the ANOVA (GLM). Thus, emphasis will be given to discussion of the results of the multinomial analyses.

In our field trials, all of the active ingredients, with the exception of *B. bassiana*, demonstrated efficacy against *C. nasturtii* by increasing the number of uninfested and marketable plants in 2013. *Beauveria bassiana* is relatively slow acting and requires direct contact with the insect host, with poor infectivity often attributed to inadequate ambient moisture (Ugine et al. 2007, Ortiz-Urquiza et al. 2010). It is likely that conditions on the apical meristem of broccoli plants, which experience direct exposure to sunlight and wind in the field, are unsuitable for successful germination of *B. bassiana* spores and for infection of *C. nasturtii* larvae. *Beauveria bassiana* appears to be a poor choice for managing *C. nasturtii*, based on ineffective field results in both years, poor oviposition deterrence, and inconsistent larvicidal activity.

Although the remaining three insecticides were all effective at reducing damage to plants in the field, on the basis of greenhouse and field trials, spinosad and pyrethrin have the greatest promise as tools for organic growers in the management of *C. nasturtii*. These two insecticides offered the most consistent results, significantly reducing oviposition for at least 24 h, causing relatively high levels of larval mortality and significantly increasing the number of uninfested and marketable plants in the field in 2013. Azadirachtin did not deter oviposition 24 h postapplication, but showed some larvicidal activity and efficacy in the field in 2013. However, the lack of field efficacy observed for these active ingredients in 2011, and seen in other field studies at similar or higher rates (Hallett et al. 2009a; Seaman et al. 2013, 2014, 2015), suggests that use of these insecticides needs to be complemented with other tactics in order to achieve consistent, effective control. Pheromone-based action thresholds can be used to time foliar applications of conventional insecticides and effectively protect cabbage from *C. nasturtii* (Hallett and Sears 2013). The applicability of this action threshold to broccoli with organic insecticides needs to be evaluated. The occurrence of oviposition-deterrent effects in these insecticides suggests that insecticide efficacy may be improved by applying foliar insecticides when gravid females are preparing to lay their eggs. However, applications occurring too far in advance of oviposition are likely to result in reduced efficacy due to environmental degradation of residual insecticide on the host plant, as spinosad, pyrethrin, *B. bassiana* and azadirachtin may be affected by temperature, UV-light, and/or moisture in the field (Fields and Korunic 2000, Williams et al. 2003, Kumar and Poehling 2006, Daniel and Wyss 2010). As such, applications of spinosad or pyrethrin should be made as soon as adult population peaks are observed through field-specific pheromone trap monitoring, in order to achieve both oviposition deterrence and maximum larval mortality.

Suppression of *C. nasturtii* damage by conventional foliar insecticides can be effective during the early phases of *C. nasturtii* colonization due to lower overall populations, but is unlikely to provide adequate control when populations are high (Hallett et al. 2009a, Chen et al. 2011). While no treatment significantly affected damage ratings at Zephyr, all treatments differed significantly from the control at Elora. *Contarinia nasturtii* abundance was not recorded in pheromone traps at Zephyr during our trial. However, a history of severe *C. nasturtii* infestations at this site and the characterization of a nearby site in Stouffville as having a “very high” population (Hallett and Sears 2013) suggest that the existence of a relatively high local population of *C. nasturtii* in Zephyr is not an unreasonable assumption. The pheromone trap data collected at Elora in 2011 (Fig. 1) is consistent with a previous characterization of this site as having a “moderate” population (i.e., ~5 males/trap/d from first emergence to late September); however, the 2013 data characterize Elora as having a “high” population (Hallett and Sears 2013). In previous studies, significant reductions in damage have been observed under “moderate” *C. nasturtii* pressure, and with no significant effects under “high” pressure (Seaman et al. 2013, 2014).

Variable efficacy from year to year may also be attributable to environmental variables such as temperature, which can affect residual insecticide concentrations, and impact local *C. nasturtii* abundance (Des Marteaux et al. 2015). The timing of transplanting seedlings into the field relative to adult population dynamics is also a contributing factor, as adult *C. nasturtii* populations fluctuate throughout the growing season. The Zephyr trial began two weeks earlier in the growing season than the Elora trial, and pheromone trap data from 2011 (Fig. 1) indicate that transplanting at Zephyr coincided with the height of the largest emergence peak of the growing season.

Since insecticides were applied on the day of transplanting, and every 7–10 d afterward, the field trials were designed to achieve continuous protection, assuming residual insecticidal efficacy of ~7 d (Hallett and Sears 2013). The high levels of damage observed at Zephyr, combined with the emergence peak occurring when seedlings were first transplanted, suggest that high local populations overwhelmed the insecticide treatments. In contrast, lower prevailing *C. nasturtii* populations during the trial period at Elora, in spite of higher overall numbers during seasonal peaks, may have been a contributing factor to the lower damage observed at that site, suggesting that the timing of transplanting may be as important a consideration when managing for avoidance of *C. nasturtii* as the characterization of the population level at a particular site. A better understanding of the residual efficacy of the insecticides evaluated herein is required in order to evaluate the applicability of the action thresholds described by Hallett and Sears (2013).

Effective management of *C. nasturtii* requires a diversified, integrated pest management strategy to control *C. nasturtii* consistently. This will be particularly true in organic systems, where available insecticides appear to be of lower efficacy than in conventional systems. The results suggest that pyrethrin and spinosad offer the greatest potential as a management tactic for growers; these are also the most accessible products for organic growers in Canada, as there are no azadirachtin pesticides registered for use on food crops in Canada. Applying these treatments at sites with lower local *C. nasturtii* abundance and timing transplants to avoid *C. nasturtii* population peaks may be key to achieving adequate levels of efficacy with these insecticide treatments. Given the in-season variability in local adult *C. nasturtii* abundance, the use of action thresholds (Hallett and Sears 2013) to guide the timing of insecticide applications could potentially result in greater levels of crop protection than the calendar-based sprays employed in these trials, since applications would be more targeted to coincide with emergence peaks. However, the action thresholds that were established based on the efficacy of conventional insecticides (Hallett and Sears 2013) would likely need to be revised in order to account for differing efficacy of the biopesticides reviewed herein (Nault and Shelton 2010). Additionally, the application of entomopathogens to reduce the number of *C. nasturtii* in the soil has shown some promise (Evans et al. 2015), and, along with regular crop rotation, may serve to complement foliar insecticides in the management of *C. nasturtii* particularly in organic production systems.

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